

Portfolio Analysis Project Assignment

Introduction

The objective of this project was to create an equal-weighted portfolio of seven publicly traded stocks and compare its risk and return characteristics against three major market ETFs: SPY, IWM, and DIA.

The portfolio required calculating individual stock factors, comparing the full portfolio against SPY, IWM, and DIA, and producing a correlation matrix between the portfolio, ETFs, and individual assets.

Data

The portfolio consists of the following seven publicly traded stocks:

Stock	Company Name	Industry
META	Meta Platforms, Inc.	Interactive Media & Services
GOOGL	Alphabet Inc.	Interactive Media & Services
DIS	The Walt Disney Company	Entertainment
BBY	Best Buy Co., Inc.	Specialty Retail
AAL	American Airlines Group Inc.	Passenger Airlines
CAT	Caterpillar Inc.	Machinery
WRB	W. R. Berkely Corporation	Insurance

The idea was to build a somewhat diversified portfolio across technology, media, retail, airlines, industrials, and insurance.

Daily close prices for the seven selected stocks and the three ETFs (SPY, IWM, and DIA) from 2016-03-24 to 2026-03-24 were downloaded using *yfinance*, then converted into daily returns for risk and performance calculations.

Methodology

The analysis began by downloading 10 years of closing prices for the seven selected stocks and the three ETFs.

There are three parts to this project:

Individual stocks vs each ETF

The first table analyzes the seven individual stocks in the portfolio.

Portfolio Weight	Each stock received the same portfolio weight of 14.29%.
Annualized volatility	Annualized volatility was calculated using the trailing 3 months of daily returns. The daily standard deviation was multiplied by square root of 252 (252 for number of trading days) - Most volatile stock – META (48.03%) - Least Volatile – WRB and AAL (20.51% and 20.86% respectively)
Beta Against SPY, IWM, and DIA	Beta was calculated as the covariance between the stock's returns and the market ETF's returns divided by the variance of the market ETF's returns. Meta had the highest beta across the ETFs. While WRB had the lowest beta with values between 0.15 and 0.38. Hence, in this portfolio, META is the most aggressive stock, while WRB is the most defensive, meaning WRB was much less sensitive to the benchmark ETFs.
Maximum Weekly Drawdown	META and CAT had the highest weekly drawdowns, 72.96% and 73.71% respectively. WRB had the smallest maximum weekly drawdown, around -41.83%, making it the most stable by this measure.
Average Weekly Drawdown	Average weekly drawdowns ranged from 27.77% to 43.86%. WRB had the least severe average drawdown, while META and CAT had the most severe drawdowns. This supports the earlier risk measures showing that WRB was relatively defensive, while META and CAT carried more downside risk.
Total Return and Annualized Return	DIS had the highest total return of about 1078.79%, and an annualized return of about 28.06%. Similarly, AAL also showed a strong total return of about 706.94%, with an annualized return of about 23.29%. META performed the worst in both metrics, with a negative total return of -72.15% and an annualized return of about -12.03%.

Portfolio vs each ETF.

Correlations	Our portfolio showed high positive correlations with all three ETFs (between 0.84 to 0.88). Hence, the portfolio moved in the same direction as the broad market ETFs.
Covariance	Ranging between 0.000138 to 0.000176, the covariance values were positive for all three ETFs, meaning the portfolio tended to move in the same direction as market ETFs.

Tracking Error	It measures how much the portfolio's returns differs from an ETF's Return. The portfolio had tracking errors between 11% to 13% against the ETFs. This indicates that although the portfolio is highly correlated with the ETFs, it still has meaningful differences from them. This makes sense because the portfolio contains only seven stocks compared to thousands for the ETFs.
Portfolio Sharpe Ratio	The Sharpe ratio measures the risk-adjusted return. The risk-free rate was assumed to be 3% annually. The portfolio Sharpe ratio was approximately 0.623. This means the portfolio generated about 0.623 units of excess return per unit of risk.
ETF Sharpe Ratio	Since the portfolio itself has only one Sharpe ratio, I also calculated the Sharpe ratio for each ETF to make the risk-adjusted comparison clearer.
Sharpe Difference	Compared to SPY, the portfolio had a slightly lower Sharpe ratio (-0.047), meaning that Spy had slightly better risk-adjusted performance than the portfolio. At 0.217 higher value than that of IWM's Sharpe ratio, the portfolio performed better than IWM on a risk-adjusted basis. Portfolio only slightly outperformed DIA on a risk-adjusted basis (0.036). Overall, portfolio's Sharpe ratio was better than IWM and DIA, but slightly worse than SPY.
Annualized Volatility Spread	Calculated as the difference of ETF volatility from Portfolio volatility, annualized volatility measures whether the portfolio is more or less volatile than the ETFs. The volatility spread was the smallest against IWM, meaning the portfolio's volatility was closest to the Russell 2000 ETF. The spread was larger against SPY and against DIA, meaning the portfolio was noticeably more volatile than those benchmarks.

Chart Interpretations

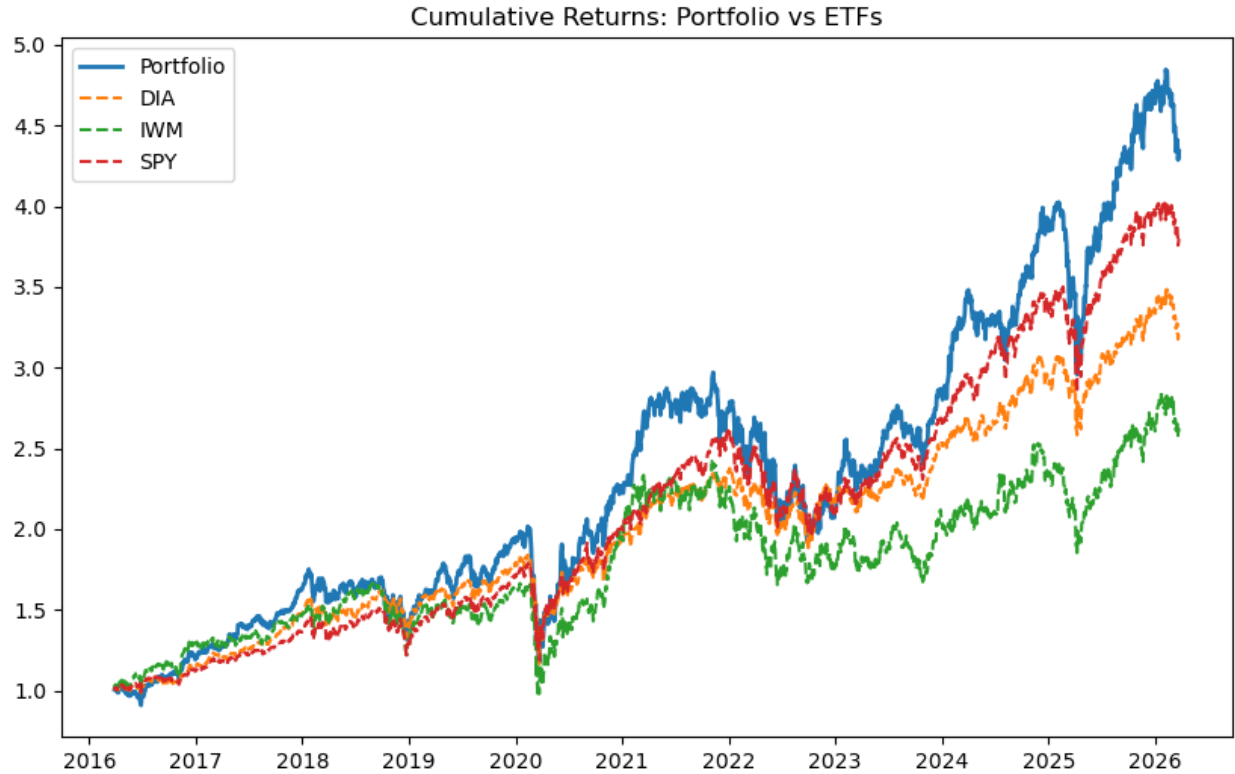


Figure 1 - Cumulative Returns: Portfolio vs ETFs

Figure 1 shows the comparison between the growth of the equal-weighted portfolio against DIA, IWM, and SPY over the full sample period. The portfolio generally moved in the same direction as the ETFs, which matches the high correlations from the portfolio comparison table. While the portfolio produced stronger long-term growth than the benchmark ETFs, the chart also highlights that our portfolio tends to have larger swings on both sides. Hence, the portfolio outperformed over time, but it also carried more volatility and downside risk than the diversified ETFs.

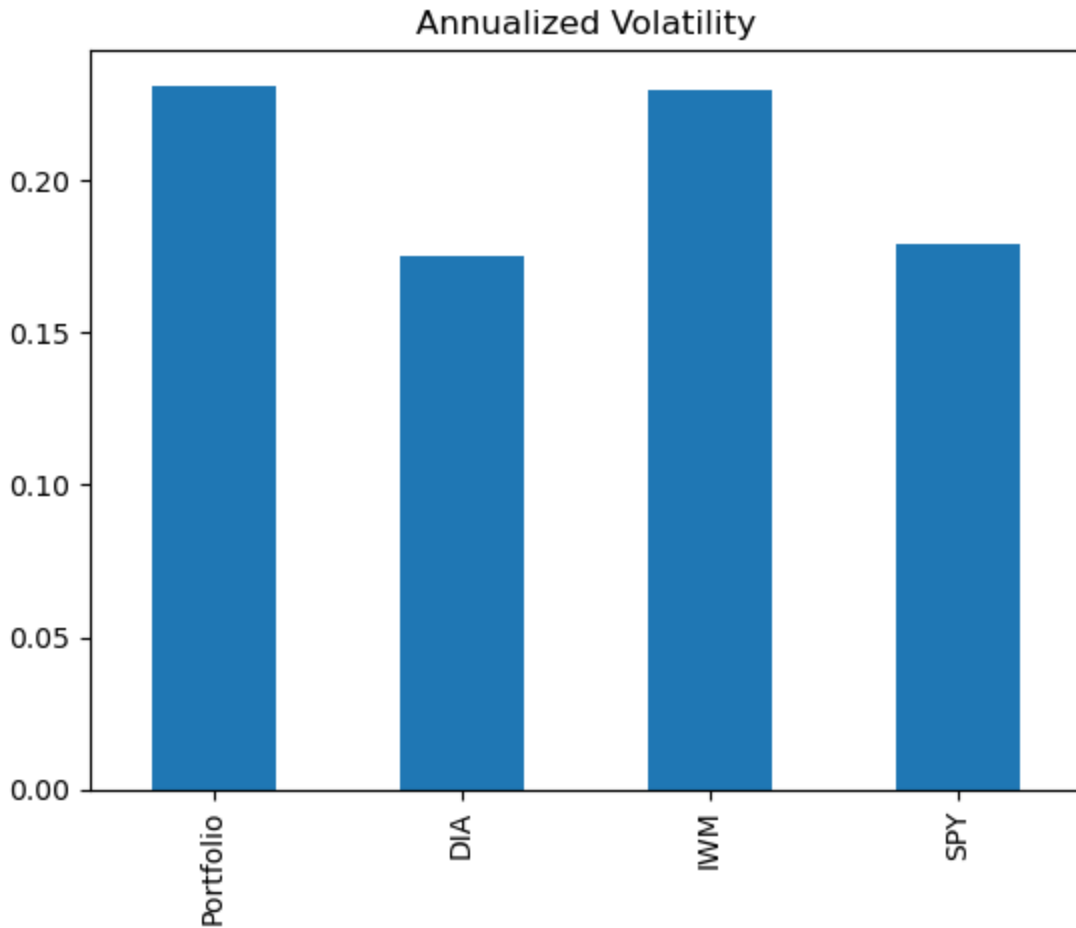


Figure 2 - Annualized Volatility Chart

Figure 2 compares the annualized volatility of the portfolio against DIA, IWM, and SPY. It shows that the portfolio was more volatile than SPY and DIA, but very close to IWM. This suggests that the portfolio's risk level was closer to the Russel 2000 ETF than to the S&P 500 or Dow Jones ETFs. Since the portfolio only contains seven stocks, it is expected to have more concentrated risk than broad ETFs.

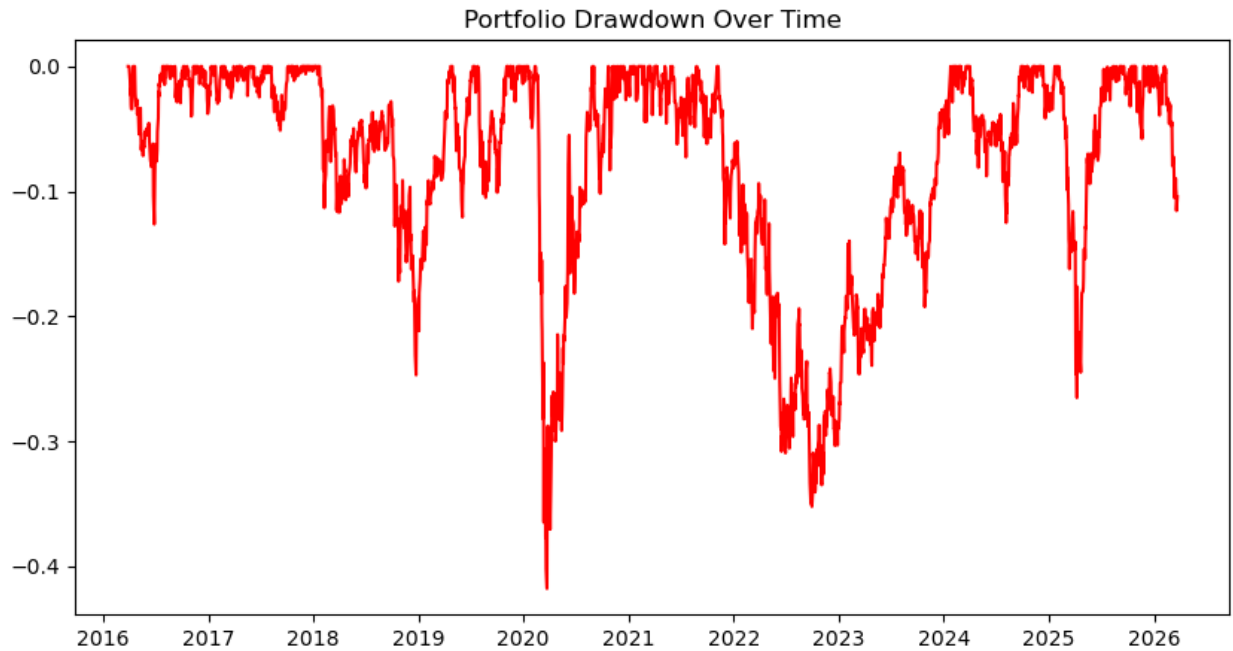


Figure 3 - Portfolio Drawdown Over Time

The drawdown chart shows that the portfolio experienced significant downside risk. The largest occurred around 2020, when the portfolio fell more than 40% from its previous peak. Another large drawdown occurred during 2022.

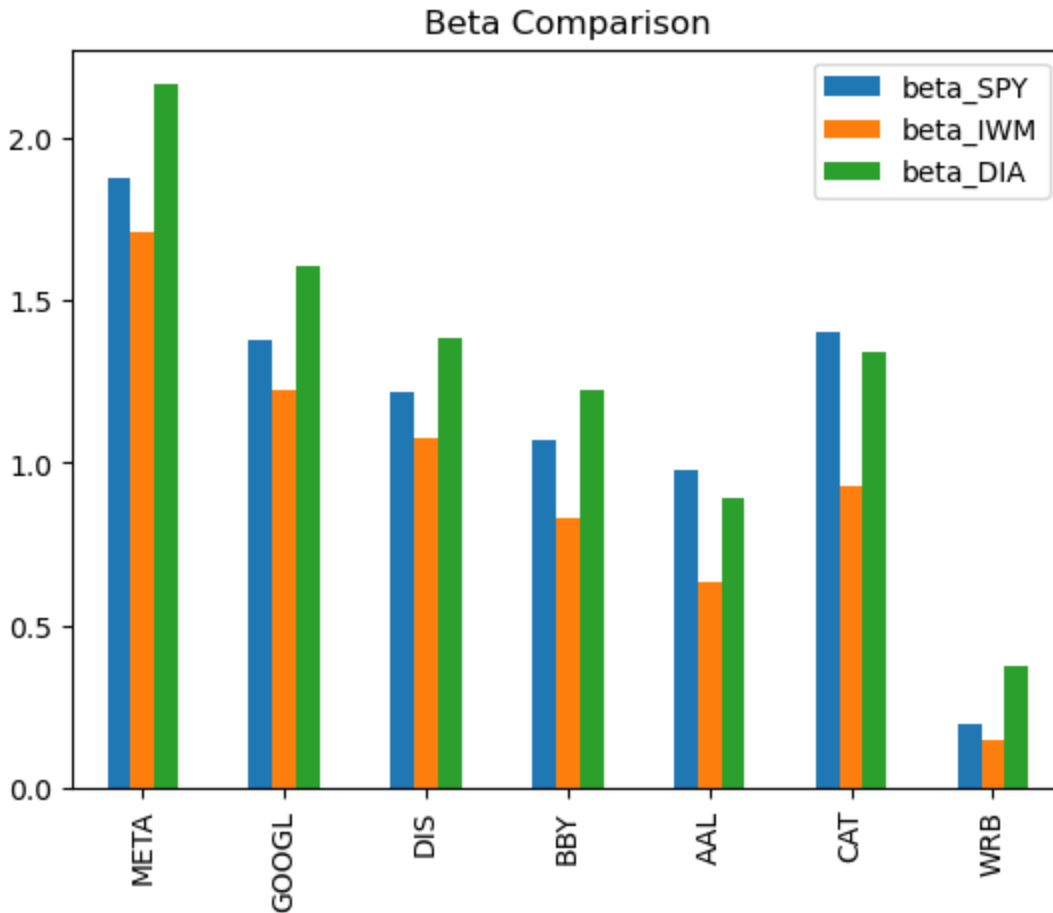


Figure 4 - Beta Comparison

Figure 4 visually confirms the beta results from the first table. META had the highest beta across the ETFs, making it the most market-sensitive stock, while WRB had the lowest beta, making it the most defensive stock in the portfolio.

Correlation matrix of individual stocks, portfolio consisting of the individual stocks, and three ETFs among each other.

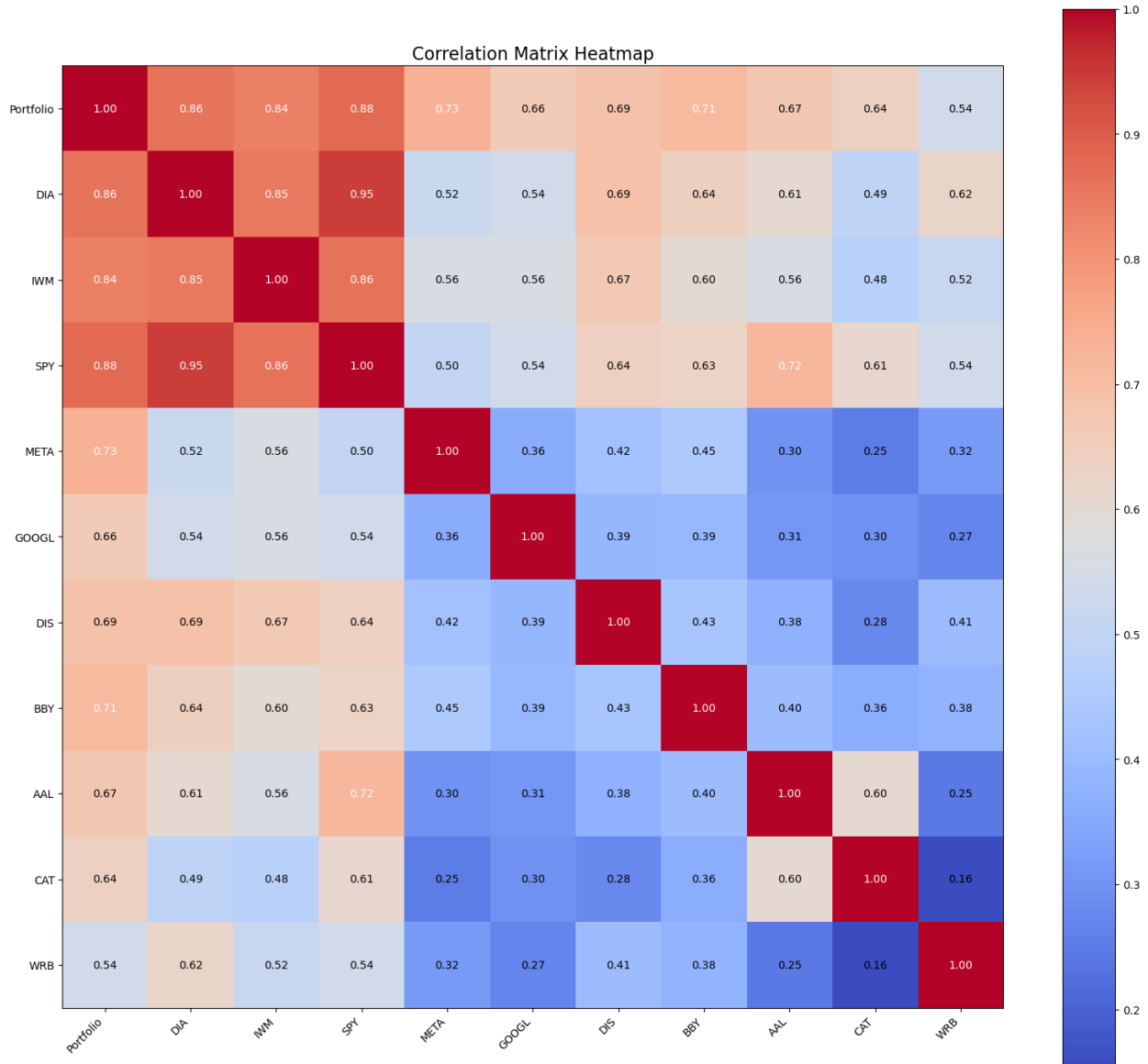


Figure 5 - Correlation Matrix Heatmap

Figure 5 shows how closely each asset moves with every other asset. The portfolio has high correlation with the ETFs. Portfolio vs DIA is 0.86, Portfolio vs IWM is 0.84, and Portfolio vs SPY is 0.88.

The 3 ETFs are highly correlated with each other with their values ranging from 0.85 to 0.95. SPY and DIA (0.95) are especially close because both are large-cap US equity benchmarks.

The individual stocks are less correlated with each other ranging from 0.16 to 0.70. Many of the stock-to-stock correlations are lower than the ETF-to-ETF correlations, which suggests that the individual stocks provide some diversification benefits. However, the portfolio is

still highly correlated with the ETFs, so it remains strongly exposed to broad market movements.

WRB has the lowest correlation with the overall portfolio, around 0.54, which supports the idea that WRB acts as the most defensive holding in the portfolio. Overall, the heatmap shows that the portfolio is strongly connected to the market, but the individual stock correlations are not so high that diversification disappears.

Conclusion

The portfolio moves strongly with the overall market, especially SPY and DIA. The portfolio had high positive correlations with all three ETFs, which shows that it generally followed market direction. With only seven selected stocks, the portfolio presented a more concentrated risk. This is reflected in the portfolio's higher volatility compared to SPY and DIA and its significant drawdowns during 2020 and 2022.

On a risk-adjusted basis, the portfolio performed slightly better than IWM and DIA based on Sharpe ratio difference, but slightly underperformed SPY. This suggests that while the portfolio produced strong long-term growth, SPY offered slightly better return per unit of risk. META was the most aggressive and volatile holding, while WRB acted as the most defensive one. Overall, the portfolio showed both market exposure and some diversification benefits, but its higher volatility and drawdowns show that it carried more risk than broad market ETFs.